

HACK4DEV IRAQ 2026
EXOPLANET DATA CHALLENGE
CHALLENGE A – TRANSIT HUNTER

Transit Hunter

A non-detection with quantified sensitivity: our photometric scatter is at or above the transit depths we were looking for.

Physics Club Team – University of Baghdad

Why This Challenge

The physics

The other challenges could be approached as data problems. This one required thinking about starlight, atmosphere and orbital motion, and turning raw telescope images into a physical measurement.

The data

We were given raw FITS frames with no labels and no feature table, so the machine-learning challenges could not be applied directly. What the data does contain is what a transit search needs: a dense time series of the same field across a night.

The reference

The planets here are already confirmed and published, so we had an independent reference to check ourselves against — which mattered more to us than the chance of finding something new.

The Problem

We look for a periodic or significant decrease in the brightness of one star across a night. A planet crossing the disc of its host removes a small fraction of the starlight, and the light curve dips.

If we look at the target star alone and see its brightness drop, we cannot be sure. It might be a planet passing in front of it, or it might be a flaw in how we measured.

A dip is a potential candidate, never a confirmed exoplanet.

WHAT MAKES IT HARD

Depths of 1.5–2.9% sit at or below the scatter a 6-inch telescope can reach.

The star field drifts during the night, so a fixed aperture slides off the star.

No flat fields and no WCS in the package.

Host stars sit at or below the detection limit at 5 arcsec per pixel.

Objective and Research Question

Can a 6-inch robotic telescope recover the known transit of an exoplanet, and what is the precision limit of our pipeline?

OBJECTIVE

Test whether our pipeline can recover a published transit signal in this data, and measure the limits of its sensitivity.

WHAT COUNTS AS SUCCESS

Either a dip that survives our own reverse test, or a quantified sensitivity ceiling that explains why no dip can be certified. Where our tests refuse to certify a candidate, we report the refusal.

Headline Result

1.08%

Photometric scatter on our best night, after detrending. Typically 4–7%.

1.5–
2.9%

Published transit depths of these planets — comparable to or smaller than our noise.

3.22σ

One marginal dip, Qatar-1 on 2026-08-21, which our own reverse test refuses to certify.

We report non-detections with quantified sensitivity. The pipeline, the diagnostics and the rejection logic are the deliverable.



PART ONE

Data and Method

Dataset

Source	MicroObservatory Image Directory	
Telescope	Cecilia, 6-inch robotic	
Total FITS files	1741 – 1681 science + 60 darks	
Targets		8
Observing sessions		22
Date range	2026-08-06 → 2026-09-05	
Image properties	650 × 500 px, 5.0 arcsec/px	
Exposure / filter / depth	60 s, Clear, 12-bit (max 4095)	

MEASUREMENTS AND VARIABLES

Each frame is a raw image. From it we derive aperture flux for the target and comparison stars, sky background from an annulus, pixel position (x, y), UT time and airmass. Relative flux — target divided by comparisons — is the variable we analyse.

The three campaign reports in this deck cover Qatar-1, TRES-3 and TRES-5. Frame counts per night are stated on each figure, and they are far smaller than the package total.

What the Package Does Not Contain

NO FLAT FIELDS

Pixel-to-pixel sensitivity is uncorrected

No flat fields exist in the MicroObservatory distribution. We partially mitigate this by detrending relative flux against (x, y) position — an approximation, not a calibration. It is a stated limitation of every result in this deck.

NO WCS IN THE HEADERS

We did not know where the target was

The RA and DEC keywords give the telescope pointing, not the target position, and the image orientation is undocumented. Solving this shaped the entire project: we plate-solved against Gaia DR3 and recovered the orientation statistically (flip-y, 585 votes against 51 and 8).

Weather is the third quality issue: 62% of frames are unsolved, mostly weather, and one entire night — TRES-1 on 2026-08-06 — has WEATHER = 0 and no detectable stars at all.

Data Preparation

Dark subtraction

60 dark frames are provided and applied. Why: removes the detector's thermal and bias signal, which otherwise varies across the night.

Saturation and invalid values

12-bit data saturates at 4095. An early rule that picked the brightest star was discarded: that star was saturated in two-thirds of frames. We use the closest unsaturated star to the field centre. Why: saturated pixels lose photometric linearity.

Differential normalization

Target flux is divided by comparison-star flux, then normalized. Why: transparency and airmass changes affect all stars together, so dividing removes them. Where no comparison survived all nights, each night is normalized to its own median — and we label it.

Sky background by median annulus

Background is taken from an annulus around each star using a median, not a mean. Why: a mean is dragged upward by faint stars that fall inside the annulus.

Frame screening

`clean_frames.py` scores all 1681 science frames on image content; `build_clean_dataset.py` copies the 5 selected nights into a clean dataset. Why: cloud-affected frames carry no usable signal — 2026-08-28 showed a median sky of 2250 counts against a typical 410.

Detrending

Relative flux is detrended against airmass and against (x, y) pixel position. Why: airmass drives atmospheric extinction; (x, y) stands in for the missing flat field. The fit is made outside the predicted transit window so it cannot absorb the signal.

The Field Drift Problem

The star field moves during the night. The telescope does not hold perfectly still, and we measured drifts of up to 75 pixels.

We measure brightness by drawing a small circle around the star and adding up the light inside it. If the circle stays fixed while the star drifts away, by mid-night we are measuring empty sky — and the brightness appears to collapse, which looks exactly like a transit.

THE OBVIOUS FIX FAILED

Searching for the brightest pixel near the star's last position sounds right. Our target is faint, so when a brighter neighbour drifted into the search box the tracking jumped onto it.

782 → 44,602

Measured flux in one frame, after the tracker locked onto the wrong star. We discarded the method.

Registration by Pattern Matching

We detect all the stars in each frame and match their geometric pattern against a reference, then move every measurement circle by the shift we recover.

WHY IT WORKS

A triangle of three stars has a shape that does not repeat by chance, so a single confused star gets corrected by the group.

WHAT IT PRODUCED

All 74 frames of that night stayed usable, and Gaia pattern-matching solved 644 of 1681 frames across the package.

WHAT WE DISCLOSE

Stacking a session at the solved position showed a real star in only 7 of 17 solved sessions, and per frame a detected star coincides with the solved target position in just 1 of 1681 frames. All photometry is therefore performed at the solved catalog position, not at an independently detected star.

Pipeline

STEP	STAGE	WHAT WAS DONE, AND WHY	OUTPUT
01	Input data	Index every FITS file in the archive so nothing is analysed by hand.	1741-row index
02	Plate solving	Match detected stars against Gaia DR3 because the headers carry no WCS.	644 / 1681 frames solved; orientation flip-y
03	Inspection	Cut out the solved position and check whether a real star sits there, rather than assuming it.	Star present in 7 / 17 solved sessions
04	Preprocessing	Dark subtraction, frame scoring, sub-pixel centroid refinement and field registration, so apertures follow the drifting field.	Clean dataset, per-frame positions
05	Window check	Propagate published ephemerides to each observing window, to know whether an event should have been there at all.	22 / 22 sessions on transit nights
06	Light curve analysis	Aperture photometry with a median sky annulus, differential normalization, airmass and (x, y) detrending.	Per-session light curves
07	Signal detection	Measure depth and scatter in the predicted window, and require a 2:1 depth-to-sensitivity ratio before certifying anything.	Depth, scatter, significance per night

The Reverse Test

A planet blocks light. It does not add light.

So we apply the same analysis to five other stars in the image and look at what happened to them at the same moment. If the target alone dipped while the rest stayed flat, the signal is real. If other stars dipped by the same amount, and some stars grew brighter, the problem is in the measurement, not in the sky.

IT ALREADY SAVED US ONCE

On one night we found a dip of 2.08%, and the published value for that planet is 2.20% — a near-perfect match. We were about to report it as a detection.

Then we ran the test and found another star in the same image that had grown brighter by more than our target had dimmed. Since a planet cannot do that, we knew our method produces numbers like these on its own, and that the match was a coincidence.

We did not report a detection. This test is what kept us from presenting a false result.

Tools

Python 3	All analysis
astropy	FITS I/O, headers, sigma-clipped statistics
photutils	Star detection (DAOSStarFinder), aperture photometry
numpy / pandas	Numerical work, tables, logs
matplotlib	All figures
Gaia DR3	Plate-solving the fields, selecting comparison stars
NASA Archive / ETD	Published transit depths and ephemerides (external source, disclosed)

NO MACHINE LEARNING

The pipeline contains no machine learning of any kind. It is classical aperture photometry using astropy and photutils, both open-source and cited.



PART TWO

Results — Three Campaign Reports

Reading the Campaign Figures

TOP STRIP — THE CAMPAIGN

Horizontal axis is real calendar date across the run, 6 August to 5 September. Each marker is one night at its true date; its height is that night's median relative flux, and the error bar is the spread of that night's measurements.

- Filled square — usable photometry
- Hollow square — measured but flagged noisy
- × Grey cross — scheduled and observed, no measurable light curve

LOWER PANELS — THE NIGHTS

Horizontal axis is UT clock time, not frame number, so gaps in the data appear as real gaps. Black points with error bars are the binned light curve; faint grey points behind them are the individual frames.

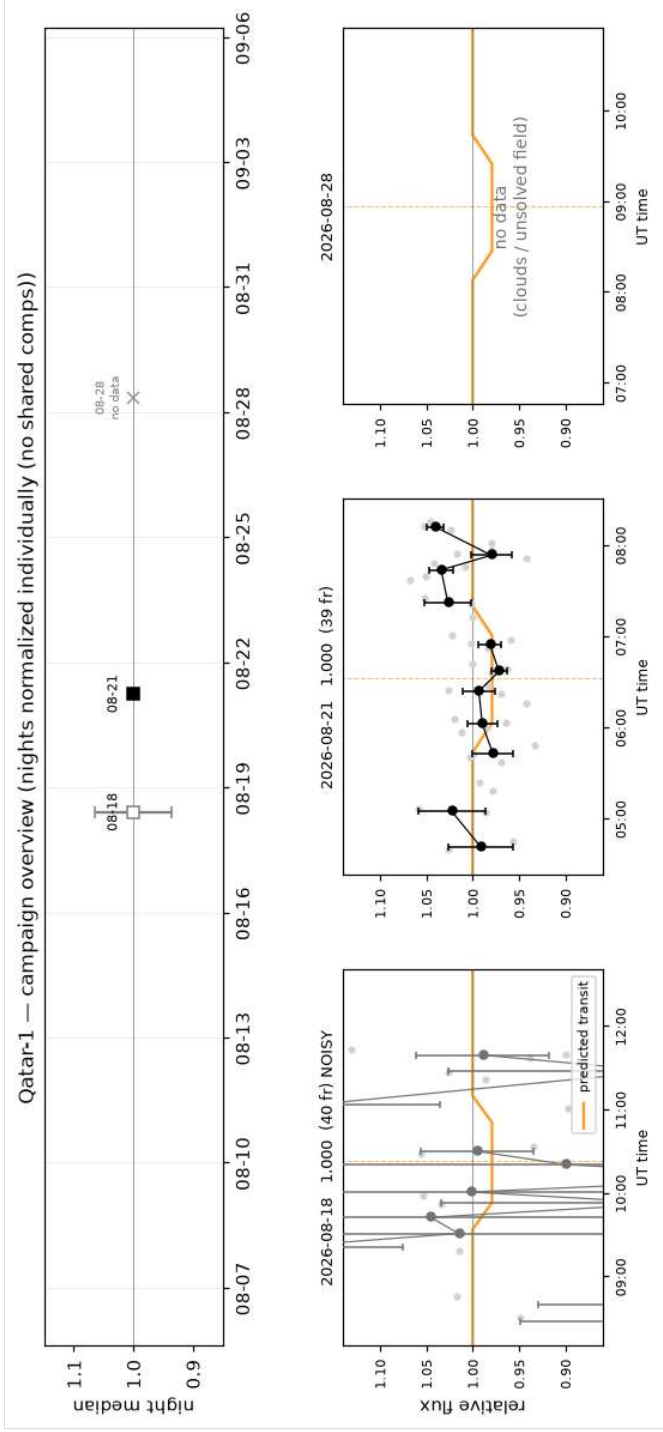
The orange line is the predicted transit model — a trapezoid built from the published depth and duration, placed at the ephemeris-predicted time. It is drawn on every panel whether or not data exist there, so you can see immediately whether our observations even covered the event.

ONE CRITICAL CAVEAT, STATED IN EVERY TITLE

Each figure states how its nights were normalized. TRES-5 reads 4 shared comps and TRES-3 reads 2 shared comps: those nights sit on a common brightness scale, so comparing them is valid. Qatar-1 reads nights normalized individually, so its top strip is not comparable between nights — the value 1.00 there is imposed by normalization, not measured.

Qatar-1

Three nights scheduled · one produced nothing at all



08-18 — NOISY

Error bars extend past 20% and frames scatter from 0.87 to 1.13. At this noise level the panel carries no information about a 2% dip.

08-21 — CLEANEST NIGHT

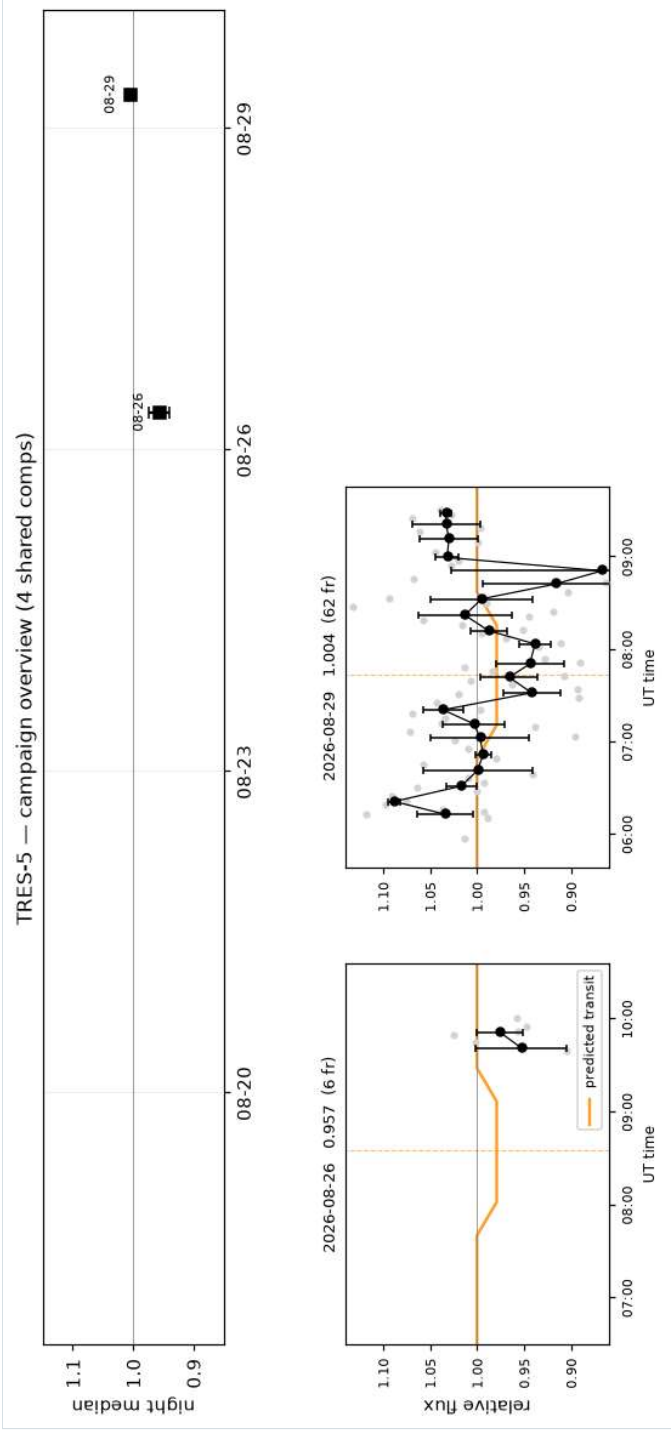
39 binned points, 04:40 to 08:20 UT, scatter around $\pm 4\%$, with a lower group between 05:40 and 07:00 reaching 0.97. The predicted window sits inside that region. This is our 3.22σ , 0.74% candidate.

08-28 — NO DATA

Clouds / unsolved field. Our independent frame-cleaning run measured a median sky of 2250 counts against a typical 410 — about five times normal. Two separate methods agree the night is unusable.

TRES-5

Our best data · 4 shared comparison stars



THE 4.7% NIGHT-TO-NIGHT STEP IS NOT REAL

08-26 sits at 0.957 and 08-29 at 1.004. A star does not change brightness by 5% between nights: 08-26 survived with only six frames, all clustered after the predicted transit had ended, so its night median is meaningless as a brightness measurement.

08-26 — THE EVENT WAS OVER FIRST

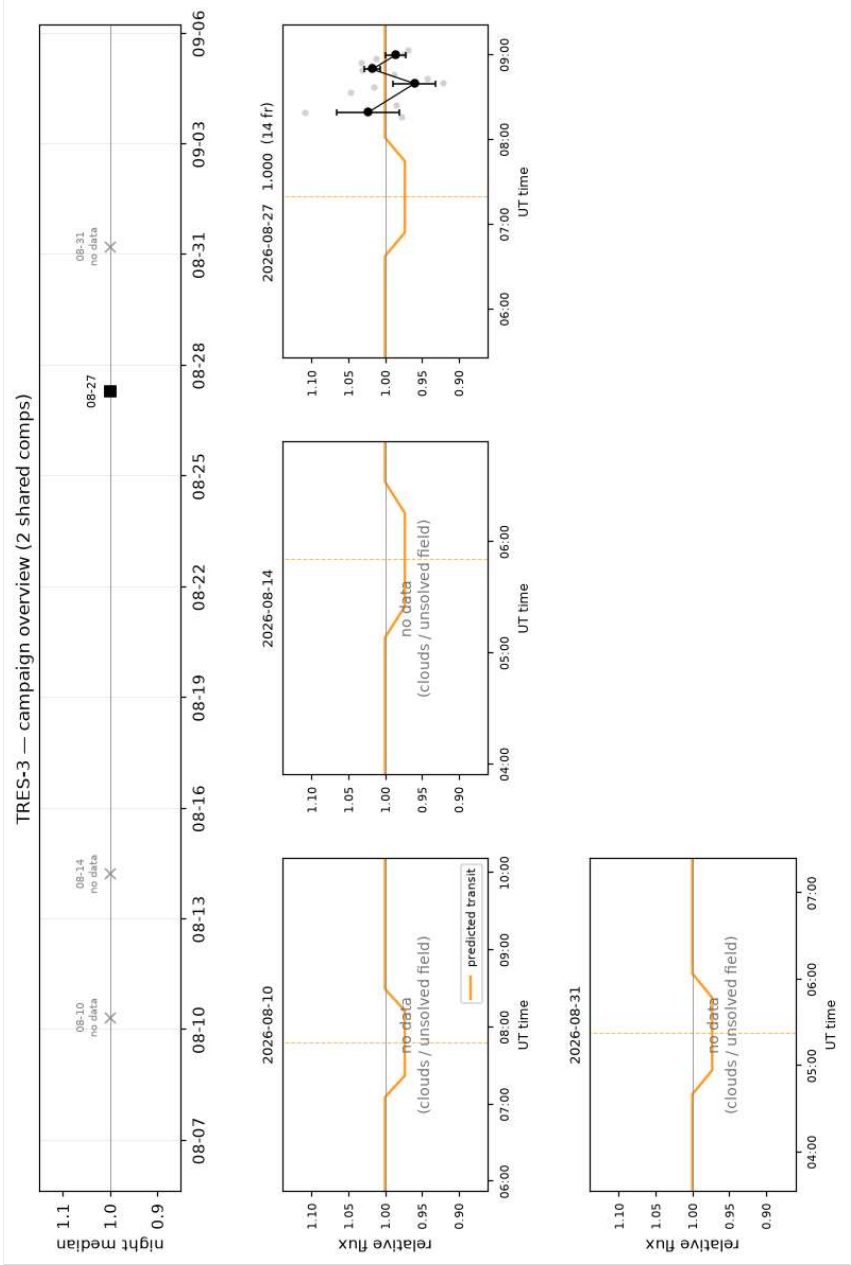
The orange model runs roughly 07:40 to 09:10 UT; our six surviving points begin at 09:30. No conclusion is possible from this night, and we draw none.

08-29 — FULL COVERAGE, INSUFFICIENT PRECISION

62 frames span 06:00 to 09:30 with the model fully inside. A low group near 07:45–08:15 reaches 0.94, close to the predicted depth — but points before the window reach 1.09 and one after drops to 0.87. Per-point error bars are 3–5% against a predicted depth of 2.2%.

TRES-3

Four nights scheduled · three produced no data at all



08-10, 08-14, 08-31 — CLOUDS / UNSOLVED FIELD

The star field could not be matched against the catalog in enough frames to build a curve.

08-27 — 14 BINNED POINTS, ALL TOO LATE

The orange model runs about 06:45 to 08:00 UT and every one of our fourteen points falls after 08:15. The transit had ended before our usable data began, and the $\pm 5\%$ scatter is in any case larger than the 2.5% depth we would be looking for.

This figure is in the submission because it produced nothing, and the reason is on the page.

Coverage, Attrition, Precision

Coverage

Of the nights that produced any light curve at all, only one — TRES-5 on 29 August — actually overlaps the predicted transit window with usable data. The orange model drawn on every panel is what makes this checkable at a glance.

Attrition

Nights that passed image-quality screening still collapsed at the photometric stage. TRES-5 on 26 August entered with 74 accepted frames and produced six; three of four TRES-3 nights produced none. Image quality and geometric registration are distinct failure modes.

Precision

Per-point error bars run 3 to 5% across all three targets, against predicted depths of 2.0 to 2.5%. The error on a single measurement is larger than the thing being measured, and no arrangement of these points can establish a detection.

The figures are the evidence for a measured sensitivity ceiling, not an illustration of it.

Why We Rejected the Candidate

QATAR-1, 2026-08-21 – OUR BEST NIGHT

Frames	74, all passing quality cuts	
Span	3.62 h	
Airmass	1.20 → 1.33	
Field drift	+39 px in x, +67 px in y	
Scatter after detrending	1.08%	
Measured depth	+0.74% at 3.22σ	
Published depth, Qatar-1b	2.0%	

1.9 : 1

Sensitivity ratio of published depth to our scatter — below the 2:1 our pipeline requires before it will certify anything.

1.8 x

Reverse test on six other stars in the same frames: the target dimmed 0.648%, a control star dimmed 0.368% — and two other stars brightened. A planet cannot brighten a star.

We report the dip as a marginal, uncertified candidate — not a transit.



PART THREE

Interpretation and Limits

The Sensitivity Ceiling

01

Every non-detection is a non-detection of an event that should have been there

Propagating published ephemerides showed that all 22 sessions overlap a predicted transit and 21 of 22 contain it fully. MicroObservatory schedules these targets on transit nights. We were not looking in the wrong place.

02

The noise floor is real; the target noise is worse

Check-star scatter — a comparison star analysed as if it were the target — reaches 0.3–3.1%, and in the best sessions falls below the expected transit depth. Target scatter is consistently several times worse. That gap is the signature of the host star sitting at or below the detection limit at 5 arcsec per pixel.

03

What the result means

With 60-second exposures from a 6-inch telescope at 5 arcsec per pixel, on host stars near the detection limit, and with no flat-field correction available, a 2% signal cannot be separated from residual systematics. This is a measured ceiling, not an assumed one.

Limitations

01 **No flat fields** in the MicroObservatory distribution. Partially mitigated by detrending relative flux against (x, y) position — an approximation, not a calibration.

03 **Ephemeris propagation error:** literature T0 values propagated over a decade give ~10–30 min uncertainty in predicted mid-transit; BJD_TDB versus UTC adds up to ~8 min. Adequate for yes/no window overlap, not for timing measurements — we make none.

05 **Orientation was determined statistically**, not from documentation. We did not force a single convention on the data.

02 **The target is measured at the solved catalog position**, not at an independently detected star, in almost all frames.

04 **5 arcsec per pixel is coarse.** Neighbouring stars blend into the apertures; contamination dilutes any real depth.

06 **62% of frames are unsolved**, mostly weather. One entire night — TRES-1, 2026-08-06 — has WEATHER = 0 and no detectable stars at all.

What We Would Do Next

BINNING

Bin in time to 10–15 minute cadence before searching for the dip. Binned check-star scatter already reaches sub-1% in several sessions and would push sensitivity below the 2% depths.

COMPARISONS

Weight comparison stars by inverse variance rather than summing equally, and require at least 4 comparisons instead of 2.

MODEL FIT

Fit a proper transit model — Mandel–Agol with limb darkening — to the binned curve with the ephemeris fixed, and report a depth upper limit per night.

FLAT FIELD

Build a master flat from the science frames themselves, using the median of many dithered frames of different fields.

INJECTION TEST

Inject synthetic transits of known depth into the data and measure how often the pipeline recovers them — turning our sensitivity estimate into a measurement.

Use of AI Tools

Where we used it

As a coding assistant while writing and debugging the Python pipeline; as a tutor for concepts none of us had worked with before — aperture photometry, sky-background subtraction via an annulus, airmass and atmospheric extinction, differential photometry, statistical significance; and as a language editor for the README.

Where we did not

No AI model is part of the solution itself. The pipeline contains no machine learning of any kind. No result, number or figure in this submission was produced by an AI model. Every value came from running our own code on the data.

What we understand

Every stage can be explained by a member of the team: why the sky annulus uses a median rather than a mean, why the airmass detrend must be fitted outside the transit window, why the reverse test is decisive, and why we report a non-detection. We did not keep any code we could not explain.

THREE SUGGESTIONS WE TESTED AND DISCARDED

Identifying the target as the brightest star in the field: it sat 235 px from the field centre and was saturated in two-thirds of frames. Replaced with the closest unsaturated star to the centre.

A local brightest-pixel tracker: it jumped onto a neighbouring star, producing a flux leap from 782 to 44,602. Replaced with three-star geometric pattern matching.

An assessment that there was no transit anywhere in the package: checked against published ephemerides, all 22 sessions were scheduled on transit nights. The assessment was wrong and we corrected it.

What We Do Not Claim

We present no feature in our results as a detected transit. Our aim was to test whether our pipeline could recover a known signal and to measure the limits of its sensitivity. Where our own tests refused to certify a candidate, we report the refusal.

EXTERNAL SOURCES DISCLOSED

NASA Exoplanet Archive / Exoplanet Transit Database – published depths and ephemerides

Gaia DR3 – field star catalogs

astropy · photutils · numpy · pandas · scipy · matplotlib

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